# Practice workbook · Find and fix errors in an algorithm

Logic and off-by-one errors in supplied pseudocode or flowchart algorithms: locating the faulty line, explaining the wrong result it causes, and writing a corrected line.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-04

# 1. Question 6 · part b.iii · 1 mark

2024 JUN · Paper 01

Worked-example retry · Attempt part b.iii only.

**6** Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

**Figure 2**

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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P 7 5 7 3 6 A 0 2 2 2 8

# 1. Question 6 · part b.iii · 1 mark

2024 JUN · Paper 01

Worked-example retry · Attempt part b.iii only.

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(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

**(1)**

(ii) Line 10

**(1)**

(iii) Line 11

**(1)**

(iv) Line 23

**(1)**

(c) Give **one** reason for the variable `index` being incremented in line 18.

**(1)**

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## 1. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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## 2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

**5** Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

**Figure 2** shows an algorithm written using flowchart symbols.

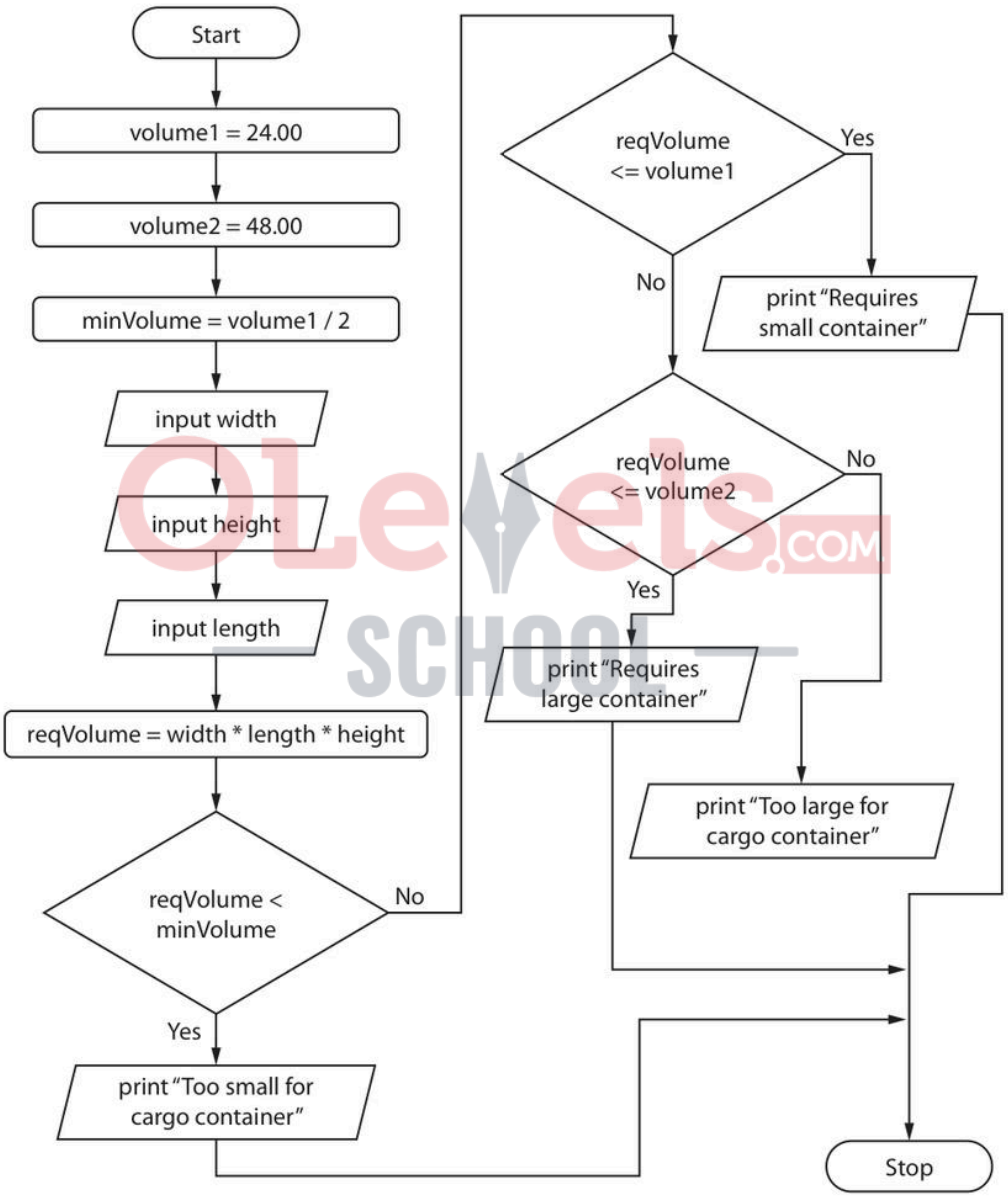



**Figure 2**

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## 2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(i) Complete the table to show the output for each cargo item.

(3)

| Cargo item | | | Output |
|---|---|---|---|
| width | length | height | |
| 4 | 4 | 2 | |
| 2 | 2 | 2 | |
| 3 | 8 | 5 | |

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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## 2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(b) Each ship is registered to a state.

**Figure 3** shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO




END FOREACH
```

**Figure 3**

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DO NOT WRITE IN THIS AREA

## 2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

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## 2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) Cargo ships have maximum weight loads.

**Figure 4** shows an algorithm written using pseudocode.

The algorithm should identify the size of cargo ship required for any load.

There is an error on line 9.

```
2   SET loadWeight TO [20000, 28000, 40000, 50000]
3   SET index TO 0
4   SET found TO FALSE
5
6   SEND "Enter cargo weight" TO DISPLAY
7   RECEIVE target FROM (INTEGER) KEYBOARD
8
9   WHILE (NOT found) DO
10      IF (loadWeight [index] >= target) THEN
11          SEND loadWeight [index] TO DISPLAY
12          SET found TO TRUE
13      ELSE
14          SET index TO  index + 1
15      END IF
16  END WHILE
17
18  IF (NOT found) THEN
19      SEND "No ship available" TO DISPLAY
20  END IF
```

**Figure 4**

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## 2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(i) Trace tables are used to identify errors in algorithms.

Complete the trace table for an input of 50500 to show what happens due to the error on line 9 in the pseudocode in **Figure 4**.

You may not need to fill in all the rows in the table.

(2)

| target | found | index | loadWeight[index] |
|---|---|---|---|
| 50500 | FALSE | 0 | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
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(ii) Construct a single line of pseudocode to correct line 9.

(2)

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**(Total for Question 5 = 12 marks)**

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## 2. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

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## 3. Question 5 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

**5** Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

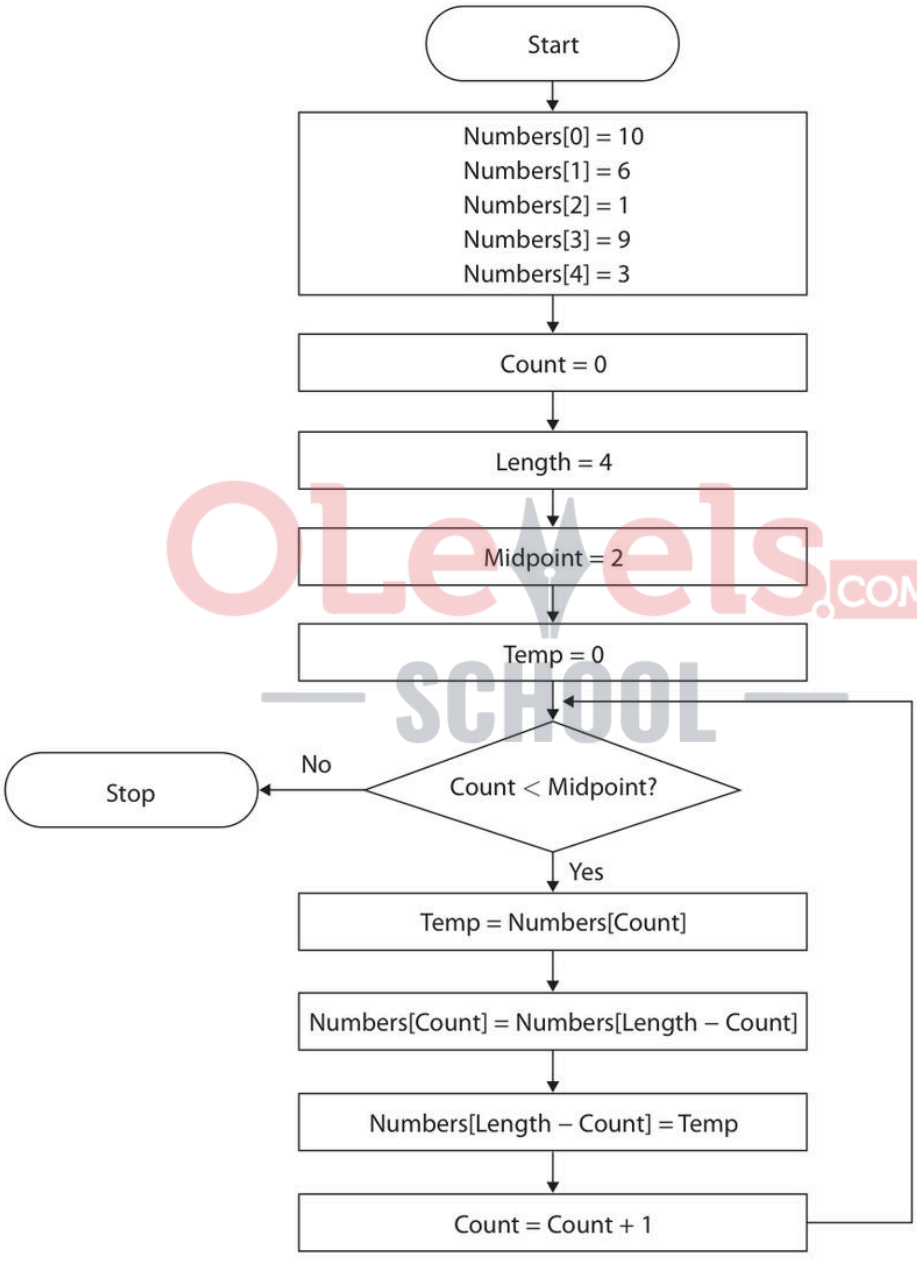



**Figure 3**

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P 6 6 4 8 9 A 0 1 2 2 0

## 3. Question 5 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

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(i) Complete the trace table. You may not need to use all of the rows.

**(5)**

| | | | | Numbers array | | | | |
|---|---|---|---|---|---|---|---|---|
| **Count** | **Length** | **Midpoint** | **Temp** | [0] | [1] | [2] | [3] | [4] |
| 0 | 4 | 2 | 0 | 10 | 6 | 1 | 9 | 3 |
| | | | | | | | | |
| | | | | | | | | |
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(ii) Give the purpose of the algorithm.

**(1)**

(iii) Explain why the variable Temp is needed.

**(2)**

# 3. Question 5 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

**Figure 4**

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

| Expected result | Actual result |
|---|---|
| The average is 2.75 | The average is 2.2 |

(i) Explain why the **Actual** result is not the same as the **Expected** result.

**(2)**

(ii) Give the number of the line that contains the error.

**(1)**

(iii) Amend a single line of pseudocode to correct the error.

**(1)**

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P 6 6 4 8 9 A 0 1 4 2 0

## 3. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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# 4. Question 5 · part b.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

**5** Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

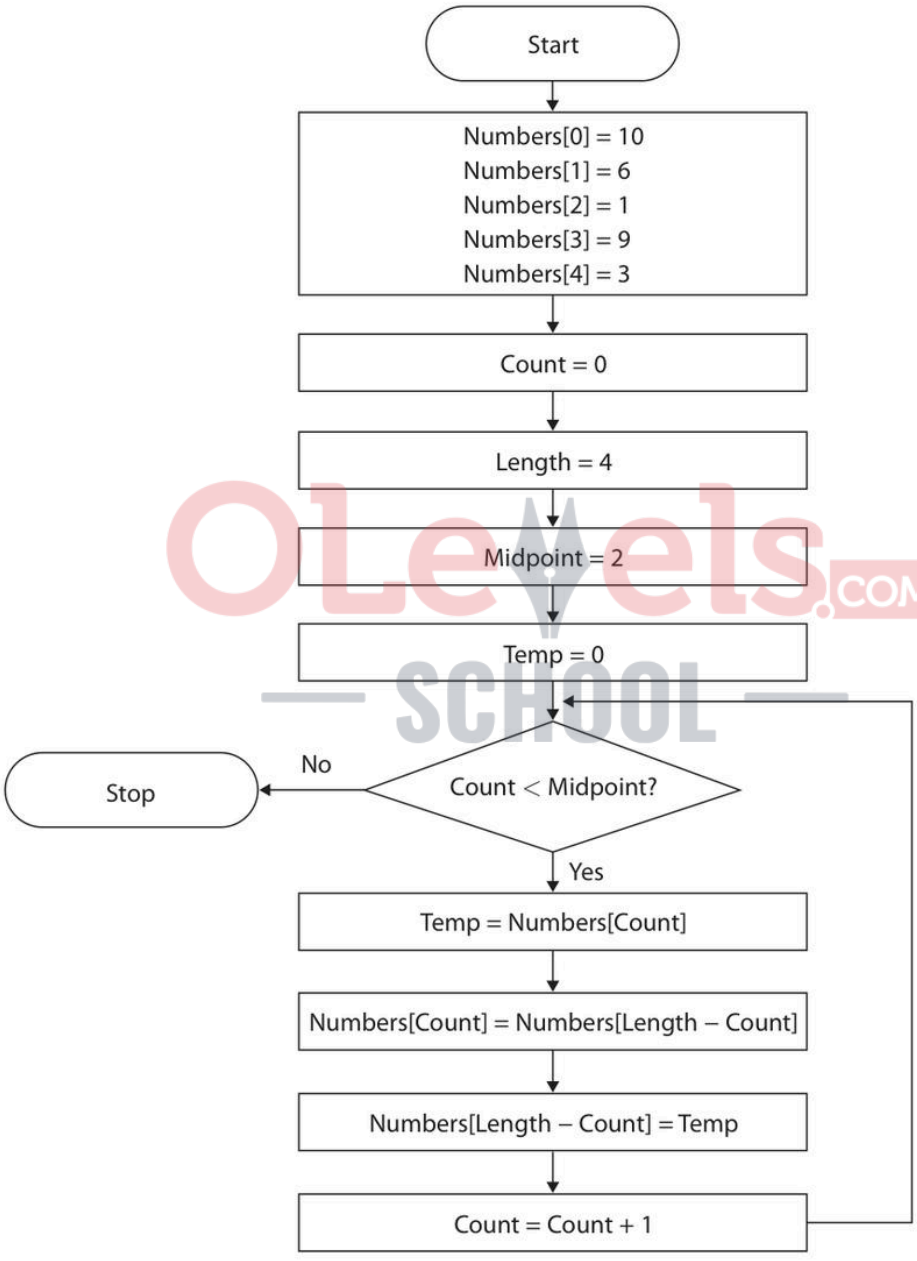



**Figure 3**

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## 4. Question 5 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

| | | | | Numbers array | | | | |
|---|---|---|---|---|---|---|---|---|
| **Count** | **Length** | **Midpoint** | **Temp** | [0] | [1] | [2] | [3] | [4] |
| 0 | 4 | 2 | 0 | 10 | 6 | 1 | 9 | 3 |
| | | | | | | | | |
| | | | | | | | | |
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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

# 4. Question 5 · part b.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

**Figure 4**

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

| Expected result | Actual result |
|---|---|
| The average is 2.75 | The average is 2.2 |

(i) Explain why the **Actual** result is not the same as the **Expected** result.

**(2)**

(ii) Give the number of the line that contains the error.

**(1)**

(iii) Amend a single line of pseudocode to correct the error.

**(1)**

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14

P 6 6 4 8 9 A 0 1 4 2 0

## 4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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# 5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.

**5** Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

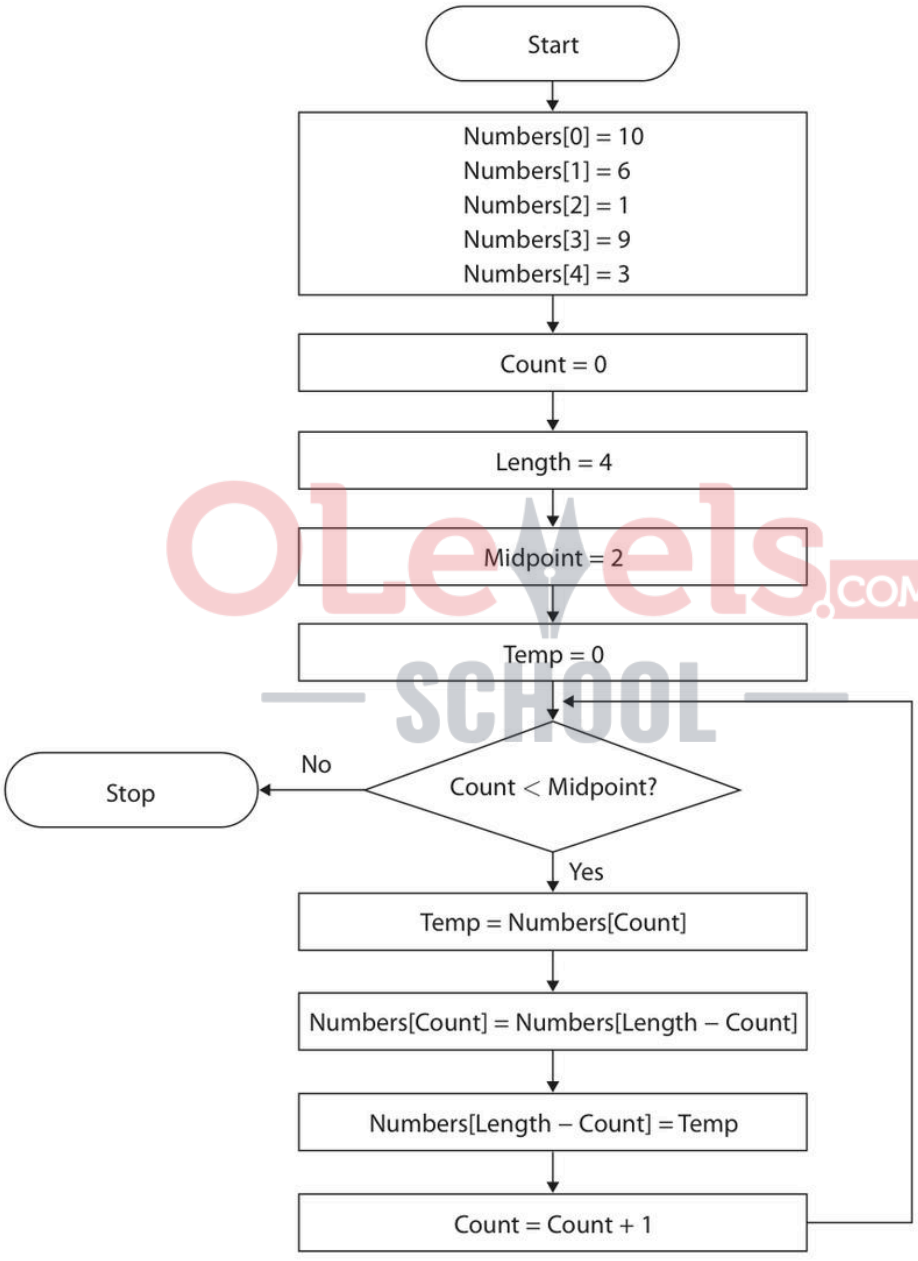



**Figure 3**

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# 5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

| | | | | Numbers array | | | | |
|---|---|---|---|---|---|---|---|---|
| **Count** | **Length** | **Midpoint** | **Temp** | [0] | [1] | [2] | [3] | [4] |
| 0 | 4 | 2 | 0 | 10 | 6 | 1 | 9 | 3 |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

# 5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

**Figure 4**

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

| Expected result | Actual result |
|---|---|
| The average is 2.75 | The average is 2.2 |

(i) Explain why the **Actual** result is not the same as the **Expected** result.

**(2)**

(ii) Give the number of the line that contains the error.

**(1)**

(iii) Amend a single line of pseudocode to correct the error.

**(1)**

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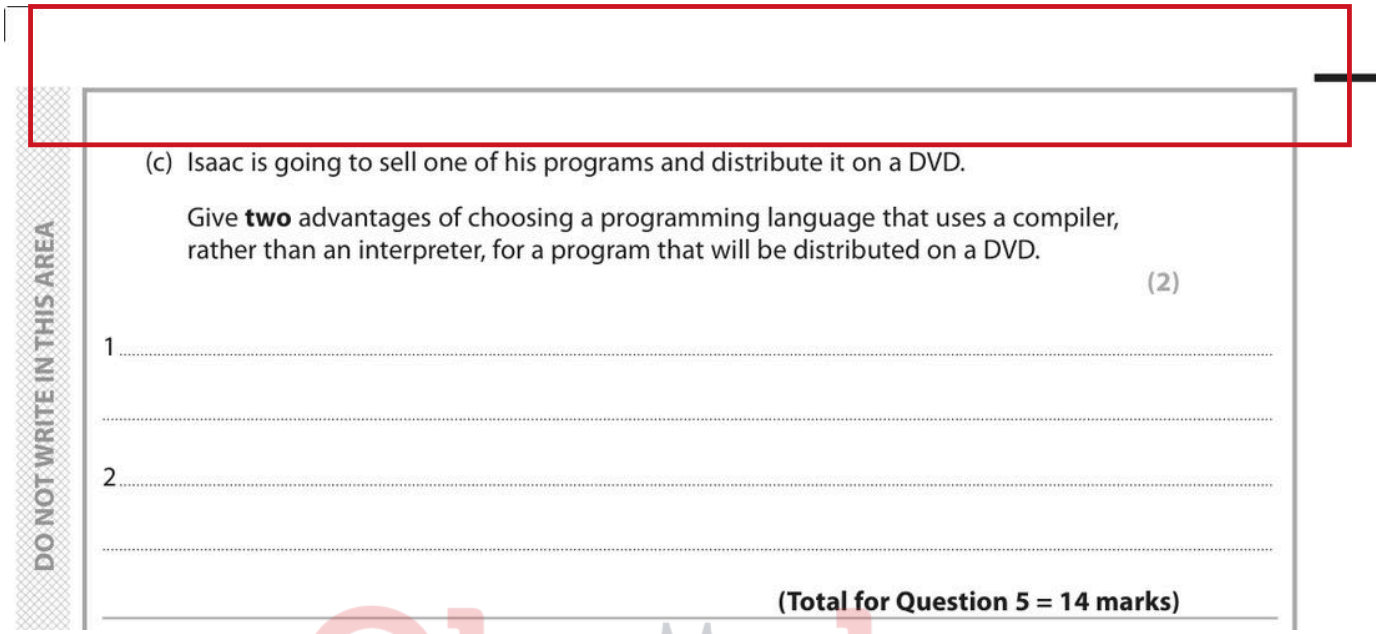
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P 6 6 4 8 9 A 0 1 4 2 0

# 5. Question 5 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.



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(c) Isaac is going to sell one of his programs and distribute it on a DVD.

Give **two** advantages of choosing a programming language that uses a compiler, rather than an interpreter, for a program that will be distributed on a DVD.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 5 = 14 marks)**

Working and answer

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# 6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

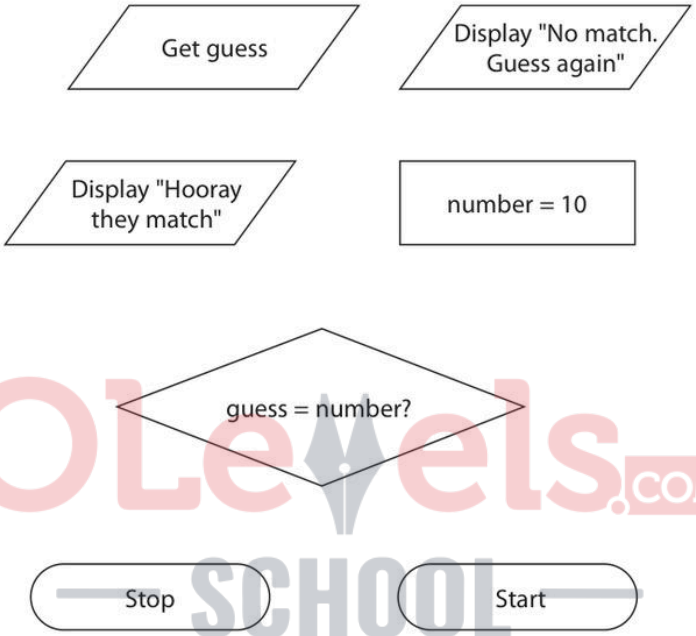
Further practice · Attempt part c.ii only.

**4** Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.




Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

**(5)**

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## 6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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**Draw your flowchart here.**

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# 6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

| | Step |
|---|---|
| A | Change the temperature to Celsius |
| B | Get the temperature in Fahrenheit |
| C | Show the temperature in Celsius |
| D | Set the temperature to 0 |

(i) Give the letter of the step that initialises a variable.

(1)

(ii) Give the letter of the step that inputs a value.

(1)

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P 7 2 5 3 8 A 0 1 2 2 0

## 6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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**QUESTION 4 (c) BEGINS ON THE NEXT PAGE.**

# 6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

**Figure 2**

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DO NOT WRITE IN THIS AREA



P 7 2 5 3 8 A 0 1 4 2 0

## 6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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Reba inputs: red, orange, red, red, orange, –1

The outputs are not as she expects.

(i) Complete the trace table to show the outputs.

(4)

| Colour | Score | RedPoints | OrangePoints | NumOranges | Outputs |
|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | |
| red | | | | | |
| orange | | | | | |
| red | | | | | |
| red | | | | | |
| orange | | | | | |
| –1 | | | | | |
| | | | | | |
| | | | | | |
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(ii) Give the line number of the pseudocode that contains the error.

(1)

.....

(iii) Write a replacement line of pseudocode to correct the error.

(1)

.....

**(Total for Question 4 = 14 marks)**

## 6. Question 4 · part c.ii · 1 mark · Working and answer

Working and answer

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# 7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

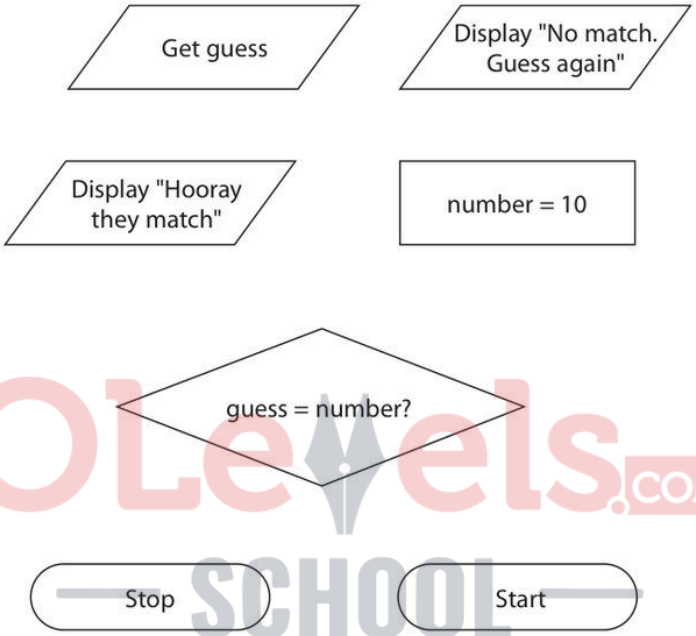
Further practice · Attempt part c.iii only.

**4** Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.




Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

**(5)**

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## 7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

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**Draw your flowchart here.**

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# 7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

| | Step |
|---|---|
| A | Change the temperature to Celsius |
| B | Get the temperature in Fahrenheit |
| C | Show the temperature in Celsius |
| D | Set the temperature to 0 |

(i) Give the letter of the step that initialises a variable.

(1)

(ii) Give the letter of the step that inputs a value.

(1)

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P 7 2 5 3 8 A 0 1 2 2 0

## 7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

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**QUESTION 4 (c) BEGINS ON THE NEXT PAGE.**

# 7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

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- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
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   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

**Figure 2**

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P 7 2 5 3 8 A 0 1 4 2 0

# 7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

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Reba inputs: red, orange, red, red, orange, –1

The outputs are not as she expects.

(i) Complete the trace table to show the outputs.

(4)

| Colour | Score | RedPoints | OrangePoints | NumOranges | Outputs |
|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | |
| red | | | | | |
| orange | | | | | |
| red | | | | | |
| red | | | | | |
| orange | | | | | |
| –1 | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |

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(ii) Give the line number of the pseudocode that contains the error.

(1)

.....

(iii) Write a replacement line of pseudocode to correct the error.

(1)

.....

**(Total for Question 4 = 14 marks)**

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## 7. Question 4 · part c.iii · 1 mark · Working and answer

Working and answer

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## 8. Question 5 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

**5** A marine scientist is conducting a study of tuna in the world's oceans.

(a) A list of tuna species needs to be sorted into **descending** alphabetical order using a merge sort algorithm.

**Figure 5** shows the lists created at the end of the splitting process.

Each list is a single tuna species.

| Bigeye | Blackfin | Albacore | Longtail | Bluefin |
|---|---|---|---|---|

**Figure 5**

It will require three passes to merge the lists into a single sorted list in **descending** order.

Complete the merge sort using the space provided.

**(2)**

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## 8. Question 5 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

(b) A bubble sort could be used to sort the list of tuna species into **ascending** order.

**Figure 6** shows an algorithm for a bubble sort.

```
1  SET myTuna TO ["Bigeye", "Blackfin", "Albacore",
2                 "Longtail", "Bluefin"]
3  SET tmp TO 0
4  SET swaps TO True
5  SET length TO LENGTH (myTuna)
6
7  WHILE (swaps = True) DO
8      SET swaps TO False
9      FOR ndx FROM 0 TO length - 1 DO
10         IF myTuna[ndx] > myTuna[ndx + 1] THEN
11             SET tmp TO myTuna[ndx + 1]
12             SET myTuna[ndx] TO myTuna[ndx + 1]
13             SET myTuna[ndx + 1] TO tmp
14             SET swaps TO True
15         END IF
16     END FOR
17 END WHILE
```

**Figure 6**

There is an error in the loop between line 9 and line 16.

Complete the table to give the line number with an error and a corrected line of pseudocode.

(2)

| | |
|---|---|
| **Line number with error** | |
| **Corrected line of pseudocode** | |

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12

P 7 2 9 3 8 A 0 1 2 1 6

## 8. Question 5 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

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(c) The scientist is collecting and storing data about tuna.

The data collected is:

- species
- length in centimetres
- weight in kilograms
- age in years.

The data is stored in an array.

The collected data needs to be written to a file named **TunaData.txt**

Each record stored in the file must have a code number in the first field.

Code numbers must start at 101

Each field in a record should be separated by a comma.

**Figure 7** shows the contents of the file.

```
101,Yellowfin,105,15,3
102,Albacore,90,15,5
103,Skipjack,50,3,4
104,Bigeye,105,25,4
105,Atlantic Bonito,50,4,2
106,Northern Bluefin,190,120,11
107,Southern Bluefin,190,120,11
108,Tongol,90,20,4
```

**Figure 7**

Open **Q05c** in the code editor.

Write the program.

You must use the structure given in **Q05c** to write the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

**(6)**

**(Total for Question 5 = 10 marks)**

## 8. Question 5 · part b · 2 marks · Working and answer

Working and answer

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# 9. Question 6 · part b.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.i only.

**6** Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

**Figure 2**

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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## 9. Question 6 · part b.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.i only.

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DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

**(1)**

(ii) Line 10

**(1)**

(iii) Line 11

**(1)**

(iv) Line 23

**(1)**

(c) Give **one** reason for the variable `index` being incremented in line 18.

**(1)**

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## 9. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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# 10. Question 6 · part b.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

**6** Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

**Figure 2**

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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P 7 5 7 3 6 A 0 2 2 2 8

## 10. Question 6 · part b.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

**(1)**

(ii) Line 10

**(1)**

(iii) Line 11

**(1)**

(iv) Line 23

**(1)**

(c) Give **one** reason for the variable `index` being incremented in line 18.

**(1)**

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## 10. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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# 11. Question 6 · part b.iv · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.iv only.

**6** Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

**Figure 2**

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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## 11. Question 6 · part b.iv · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

**(1)**

(ii) Line 10

**(1)**

(iii) Line 11

**(1)**

(iv) Line 23

**(1)**

(c) Give **one** reason for the variable `index` being incremented in line 18.

**(1)**

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## 11. Question 6 · part b.iv · 1 mark · Working and answer

Working and answer

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# 12. Question 5 · part a.i · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

**5** Martha is a computer science student and is learning to code.

(a) **Figure 7** shows an algorithm in pseudocode written by Martha.

```
1  SET first TO 0
2  SET second TO 0
3  SET answer TO 0
4  SET user_answer TO 0
5  SET score TO 0
6
7  REPEAT 10 TIMES
8      SET score TO 0
9      SET first TO RANDOM(99)
10     SET second TO RANDOM(99)
11     SET answer TO first + second
12
13     SEND first "+" second "=" TO DISPLAY
14     RECEIVE user_answer FROM (INTEGER) KEYBOARD
15
16     IF user_answer = answer THEN
17        SET score TO score + 1
18     END IF
19 END REPEAT
20 SEND score TO DISPLAY
```

**Figure 7**

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DO NOT WRITE IN THIS AREA
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## 12. Question 5 · part a.i · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

There is a logic error on line 8 of the pseudocode.

(i) Explain the problem this error will cause.

**(2)**

(ii) Describe the purpose of the algorithm shown in **Figure 7**, assuming the error has been corrected.

**(2)**

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## 12. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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